Predicting hydropower unit commitment in Tokke–Vinje: from feature engineering to a first-principles optimiser

TDT4173 Modern Machine Learning in Practice — SINTEF task — Group 300

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Abstract

We predict, for every hour of a one-week scheduling case and each of the 14 generating units of the Tokke–Vinje hydropower system, the probability that SINTEF's SHOP optimiser commits the unit. The work followed three tracks. *Track A* engineered features for gradient-boosted trees — economic spreads between price and water value, terminal water values, a count model over hours-on per unit-week, hourly block structure, a long (case, hour, unit) format — reaching 0.9638 micro ROC-AUC in walk-forward validation and 0.98058 on the public leaderboard. *Track B* starts from the observation that the labels are the output of a known optimisation problem whose inputs we hold in full: we rebuilt SHOP's weekly problem as a linear programme from the SHOP-ready topology file and use its solution *only as features* for LightGBM, which scores 0.9912 in walk-forward validation, improving every fold and all 14 units over Track A, and 0.99522 on the public leaderboard. *Track C* trains a sequence neural network — temporal convolution, a bidirectional GRU along each unit's week and attention across units — on 83 base features and 16 closed-form physics features derived from the generator efficiency curves, with no solver; it scores **0.99024** on the public leaderboard. Tracks B and C are our two selected predictions. Four predictor types compared on identical features show that Track B's gain comes from the optimiser features rather than from the learner.

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1 The task and the data

Short-term hydropower scheduling decides, hour by hour, which generating units run and at what level. SINTEF's SHOP software solves this as a mixed-integer optimisation problem; the InterOpt project asks whether a machine-learning model can predict SHOP's on/off (unit-commitment) decisions from the inputs, to speed up or warm-start the optimiser.

The Tokke–Vinje watercourse has 17 reservoirs, 9 plants and 14 units (1,000 MW installed). Each *case* is a 168-hour week starting at 00:00 UTC; cases start every day, so consecutive cases overlap by 144 hours. There are 2,922 labelled cases (2015-01-01 to 2022-12-31) and 725 test cases (2023-01-01 to 2024-12-25). The model must output, for each test case, unit and hour, a value in $[0, 1]$: the probability that the unit is on. The metric is micro-averaged ROC-AUC over every (case, hour, unit) cell — 6.9M labelled cells and 1.7M test cells — so probabilities must be comparable across units as well as ordered within each unit-week.

file	resolution, coverage	content
core		
`Historical_day_ahead_price`	hourly, 2015–2025	day-ahead price, EUR/MWh
`Historical_inflow`	hourly, 1958–2025-10	inflow to the 17 reservoirs and one river, m^3/s
`Historical_volume`	daily, 2015–2025-01-01	storage at the start of each day, Mm^3
`Synthetic_water_value`	daily, 2015–2025-01-01	marginal water value per reservoir, EUR/MWh
`Unit_commitment_decisions`	per case, 2015–2022	labels: on/off for 14 units × 168 hours; SHOP solve time
extended (optional)		
`Constraint_min_volume`	hourly, 2015–2025	minimum storage (3 reservoirs), Mm^3
`Constraint_min_flow`	hourly, 2015–2025	minimum flow (6 rivers), m^3/s
`Tokke_Vinje_topology.yaml`	static	SHOP-ready system: reservoirs, rivers, tunnels, plants, generators
`Tokke_topology.pdf`	static	diagram of the watercourse (Figure 1)

Table 1: The data. All feature files cover the whole test period; labels stop at 2022-12-31.

Three tracks. The work ran in three tracks, and the report follows them through every section. *Track A* (Peter Klivnoy) engineered features for gradient-boosted trees and produced our first submission (0.98058). *Track B* (Peter Klivnoy) rebuilt the optimisation problem that generated the labels as a linear programme and uses its solution as features for LightGBM. *Track C* (Anders Bærheim) built closed-form physics features from the generator curves and a sequence neural network over each case. Tracks B and C are the two selected predictions (Short notebooks 1 and 2).

2 Exploratory data analysis

Five rounds of EDA shaped every later decision. This section is organised by the six activities in the course's EDA list; we did all six.

2.1 Search domain knowledge

We read the task statement, the dataset documentation and the SHOP-format topology file (whose attributes are defined in the SHOP documentation), and used the standard economics of hydropower scheduling. The facts that mattered:

- **Water value is an opportunity cost.** Water released now cannot be sold later. A unit is worth running when the price exceeds the value of the water it uses, the reservoir's *water value* (EUR/MWh). The dispatch condition is therefore the *spread* between price and water value, not the price alone.
- **The week has a boundary condition.** A weekly schedule values the water left at the end of the week at the *terminal* water value; the documentation states which day's value that is.
- **Starting a unit costs money** (start/stop costs in the topology file), so units run in contiguous blocks rather than hour by hour.
- **The system is a cascade.** Water released upstream flows into downstream reservoirs and plants (Figure 1); a plant's decision depends on the reservoirs above and below it.
- **Identical units share a penstock** at Tokke (four units) and Vinje (three), so which one starts first is a matter of hydraulic losses and tie-breaking, not of different machines.
- **Norwegian inflow is dominated by snowmelt** in May–June; reservoirs are drawn down through winter and refilled in summer.

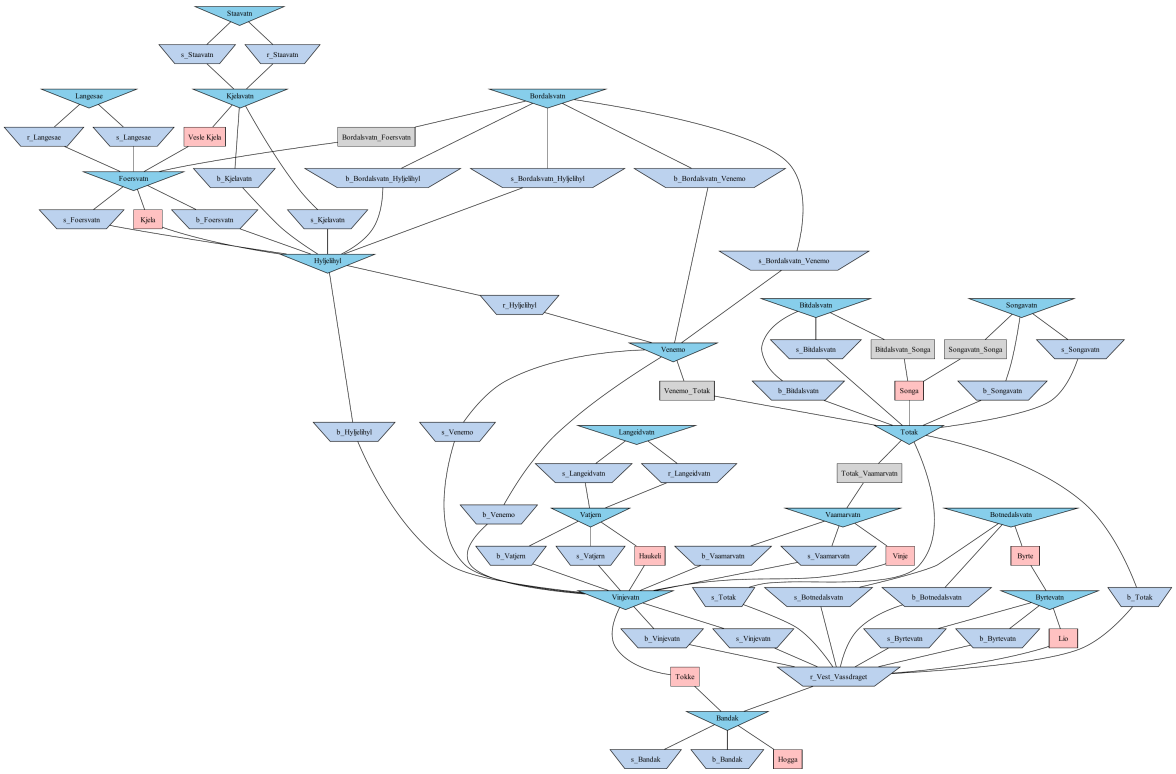


Figure 1: The Tokke–Vinje watercourse, as provided with the data (`Tokke_topology.pdf`): reservoirs (triangles), rivers and spillways (trapezoids), tunnels (grey) and the 9 plants (red). Water flows down the page to Bandak.

plant	units	$p_{\min}$ (MW)	$p_{\max}$ (MW)	start/stop cost (EUR)
Tokke	G1–G4 (identical)	36.9	107.5	214
Vinje	G1–G3 (identical)	34.4	100.0	198
Songa	G1	41.2	120.0	222
Kjela	G1	20.6	60.0	171
Lio	G1	14.8	43.0	163
Byrte	G1	6.9	20.0	135
Hogga	G1	3.0	17.0	94
Vesle Kjela	G1	2.9	8.5	92
Haukeli	G1	0.3	4.9	143

Table 2: System configuration from `Tokke_Vinje_topology.yaml`: 14 units, 1,003 MW in total. Tokke and Vinje hold 73% of the capacity.

2.2 Understand how the data was generated

- **The labels are optimiser output.** Every case was solved by SHOP v17.17 on the Tokke–Vinje model described by the topology file. The labels are therefore a deterministic function of the inputs we hold, not noisy observations of the real plant.
- **Cases are rolling.** A new case starts every day, so consecutive cases share 144 wall-clock hours, solved from almost the same state. The same hour solved one day later gets a *different* decision 12% of the time (18% for Tokke and Vinje; Figure 2): the decision depends on the whole horizon, including the end-of-week boundary, not only on the hour. This label self-disagreement explains 78% of the variance in per-unit difficulty of our first baseline. It also means random row splits leak (Section 3).
- **The water values are synthetic** (file name and documentation), one per reservoir per day. For a case starting on day d the documentation defines the initial value at d and the final (terminal) value at $d + 7$ 00:00. We first read the terminal value one day late and found the off-by-one by rereading the documentation.
- **Two storage series are invented:** Hyljelihyl and Vatjern have no historical records and are set to 50% full throughout.
- **The storage series is observed, not planned:** it is the reservoir level at the start of each day, so the level at the end of a case reflects decisions taken during it ("post-decision" information, Section 2.6).
- **SHOP's conventions can be recovered from the labels.** Rebuilding the generator (Track B, Section 4.2) required three conventions the documentation does not state — how water values are converted to a value per cubic metre, how river costs are read, how identical units are ordered — and we measured each of them from the labels (Figure 16).

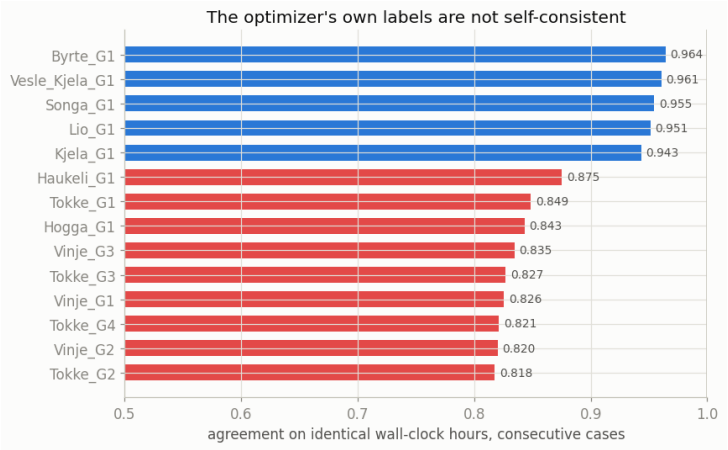


Figure 2: Agreement between consecutive cases on identical wall-clock hours, per unit.

2.3 Check if the data is intuitive

Most series behave as domain knowledge predicts. Inflow peaks in May–June at about 8 times the winter level, and storage follows the expected annual fill and drawdown (Figure 3). The system water value tracks price but is much smoother, as an expected *future* value should be. Commitment has a double-hump daily pattern (trough 0.46 at night, peaks 0.55 at the morning and evening ramps) and almost no weekday pattern (Figure 4).

One fact is counter-intuitive: **2022 has the highest prices (∼211 EUR/MWh) and the lowest on-rate (0.34)** (Figure 5, left), and raw price correlates *negatively* with commitment (−0.19). The resolution is in the right panel: water values rose with prices, so high prices did not mean a high *spread*; 2022 was a scarce-water year in which SHOP conserved water. Measured within each week, price correlates positively (+0.26 for the within-week z-score). The data is consistent once read through the opportunity cost of water.

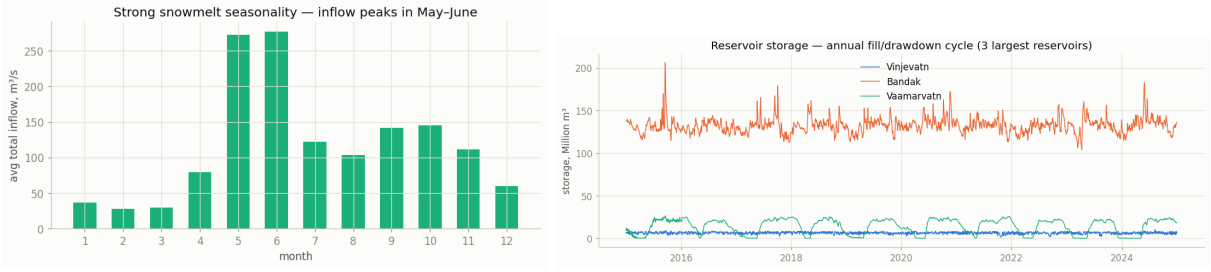


Figure 3: Left: mean total inflow by month (snowmelt in May–June). Right: daily storage of the three largest reservoirs.

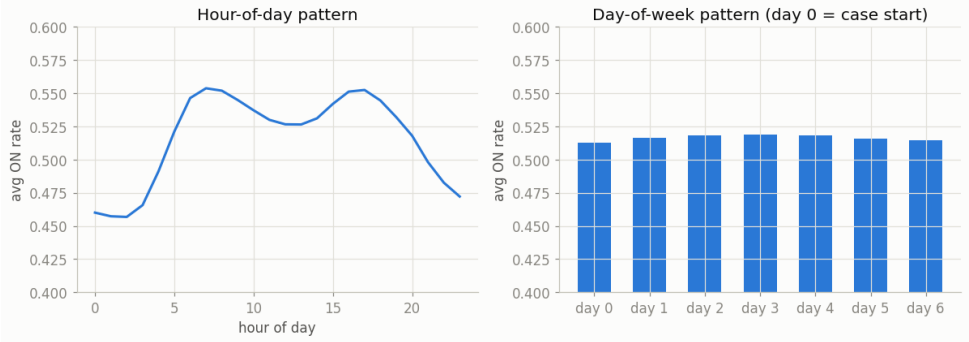


Figure 4: On-rate by hour of day and by day of week.

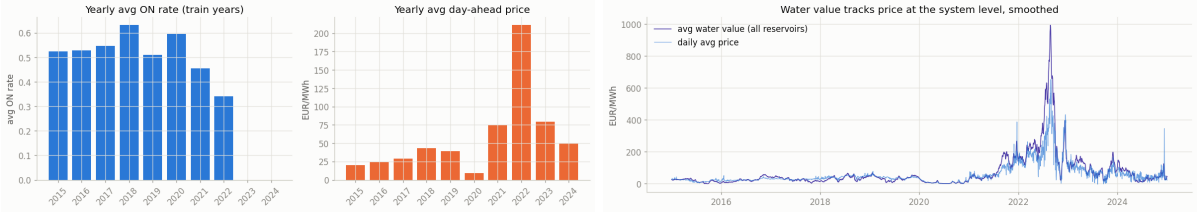


Figure 5: Left: yearly on-rate against yearly mean price — the highest-price year has the lowest on-rate. Right: system water value against price; the two rise together, so the spread does not.

2.4 Explore individual features

Target. On-rates range from 26% (Vinje G3) to 75% (Lio G1): fourteen differently balanced problems pooled by one metric (Figure 6, left). Most unit-weeks are flat: Haukeli never switches in 81% of weeks, Vesle Kjela in 75%, Byrte in 74% (right); within a week commitments come in 1.2–2.7 contiguous blocks, $P(\text{on}_t \mid \text{on}_{t-1}) = 0.995$. The units are not stationary: Haukeli's on-rate falls from 0.87 (2015) to 0.14 (2022) and Songa swings $0.30 \to 0.71 \to 0.12$ (Figure 7).

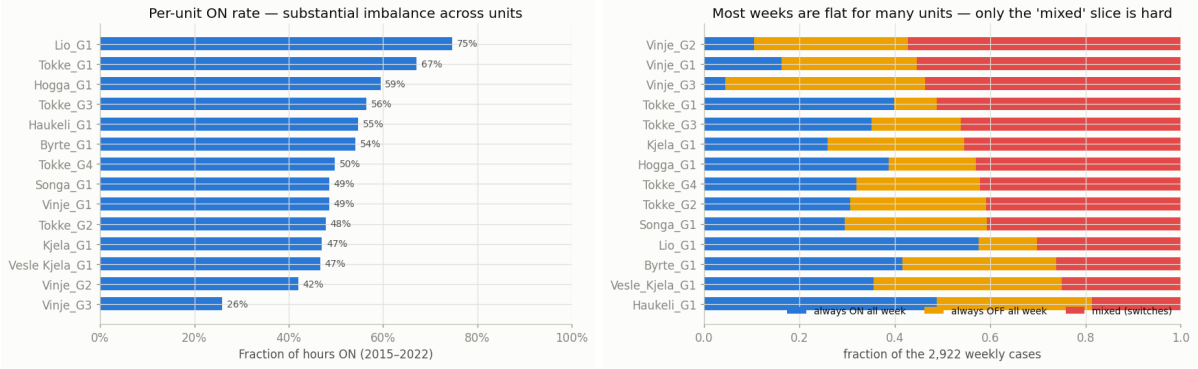


Figure 6: Left: on-rate per unit, 2015–2022. Right: share of unit-weeks that never switch.

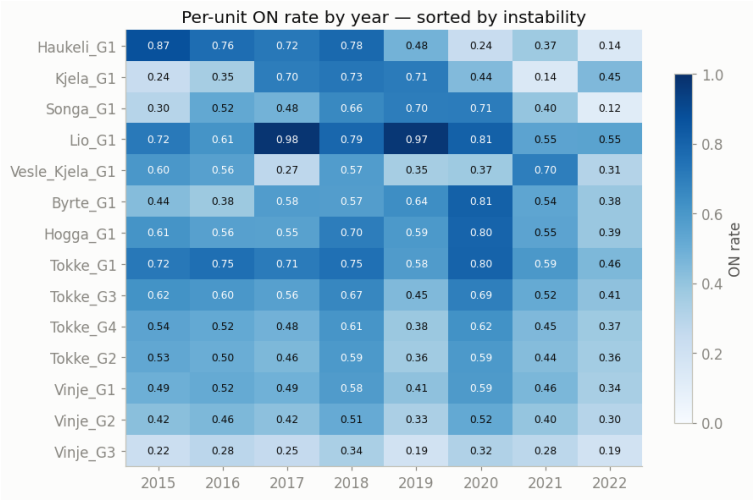


Figure 7: On-rate per unit by year, sorted by instability.

Price. Stable at 9–43 EUR/MWh until 2020, then the 2021–22 energy crisis; the test period 2023–24 sits between the two regimes and contains negative prices in 2.2% of hours (Figure 8).

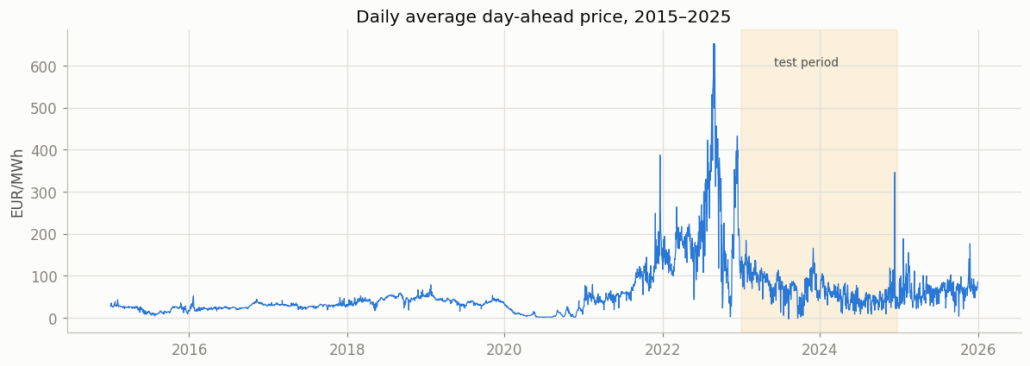


Figure 8: Daily mean day-ahead price, test window shaded.

Inflow, storage, water value. Strongly seasonal (Figure 3). Raw levels are weak predictors because they confound the season: every reservoir is full in June. Normalised against the same time of year, they become the strongest univariate correlates in the project — storage's seasonal percentile +0.54, inflow's trailing-year mean +0.33 (Figure 13, Section 2.5).

Constraints. Minimum-volume constraints bind for exactly three reservoirs and only from July to October. Minimum flow is positive in every hour for five of the six constrained rivers, so a binary "constrained" flag would be constant; the information is in the level and its changes.

2.5 Explore pairs and groups of features

Units together. Units at one plant correlate 0.8–0.96, and Tokke and Vinje correlate 0.8–0.9 with each other because Vinje's discharge feeds the reservoir Tokke draws from (Figure 9, left). Identical units follow a near-strict merit order (right): one Tokke unit on is G1 99% of the time, two are G1+G3 94%; knowing how many units are on predicts exactly which ones with 97.8% (Tokke) and 99.9% (Vinje) accuracy.

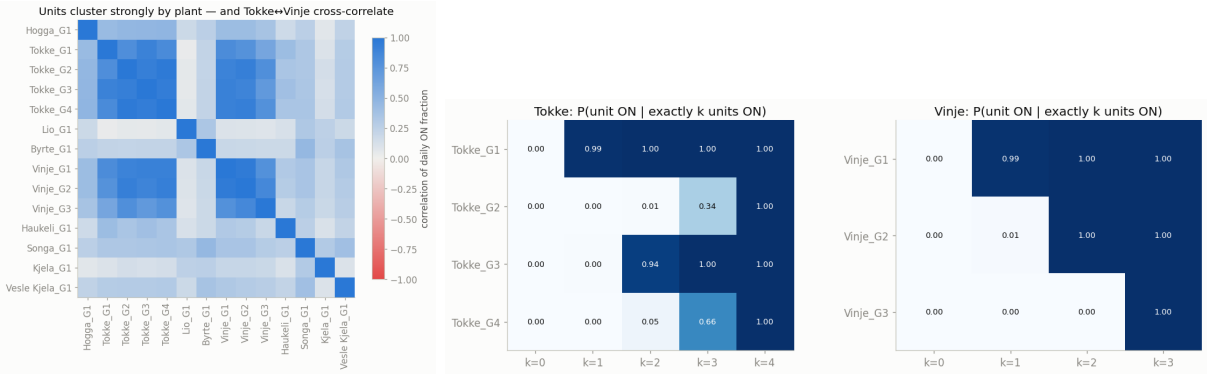


Figure 9: Left: correlation of the units' daily on-fraction, blocks by plant. Right: P(unit on | k units on at the plant) for Tokke and Vinje.

Price with water value. The spread price − water value of the feeding reservoir correlates with commitment more strongly than either component, for every unit (Figure 10, left). Price helps only relative to its week: subtracting the case mean flips its correlation from −0.185 to +0.183 and dividing by the case's standard deviation gives +0.263 (right). The spread must *not* be normalised the same way (+0.241 raw, +0.114 z-scored): it is already relative.

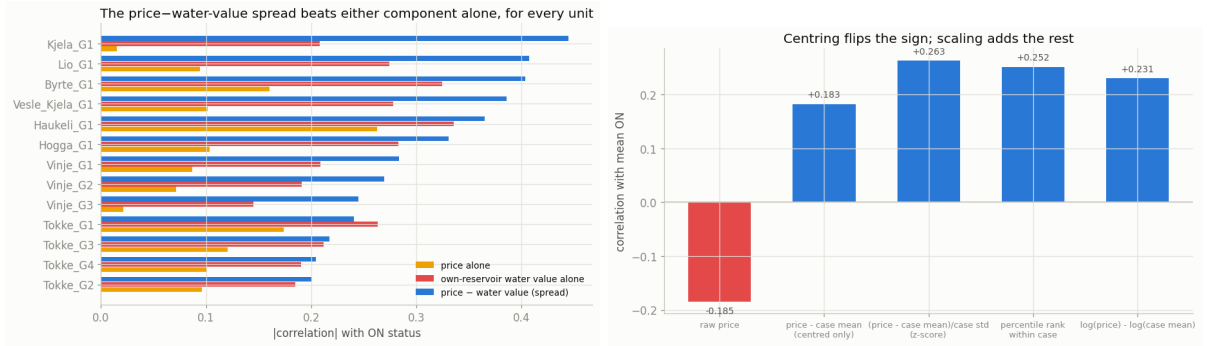


Figure 10: Left: |correlation| with commitment of price, own water value and their spread, per unit. Right: correlation of five price transformations with commitment.

Groups of series. The 17 water-value columns share one factor (PC1 explains 91% of their variance); inflow needs 3 components for 90%, storage 7 (Figure 11). The useful decomposition is a system-wide level plus each reservoir's deviation from it; the deviation is the strongest of the three for five units but useless for Tokke, whose reservoir tracks the system.

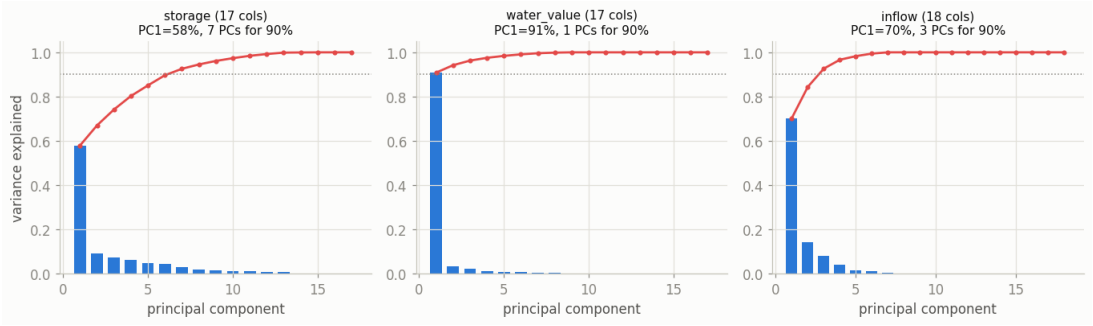


Figure 11: Variance explained by principal components within each family of series.

Across time. The terminal water value correlates −0.41 with case-level commitment against −0.26 for the initial value, and the two differ in 93% of case/reservoir pairs (Figure 12, left). Price correlates with commitment most weakly at the current hour and more strongly at leads and lags of several days (right): SHOP plans the whole week. Normalising the slow series against the same time of year raises their correlation 1.3–2.6× (Figure 13).

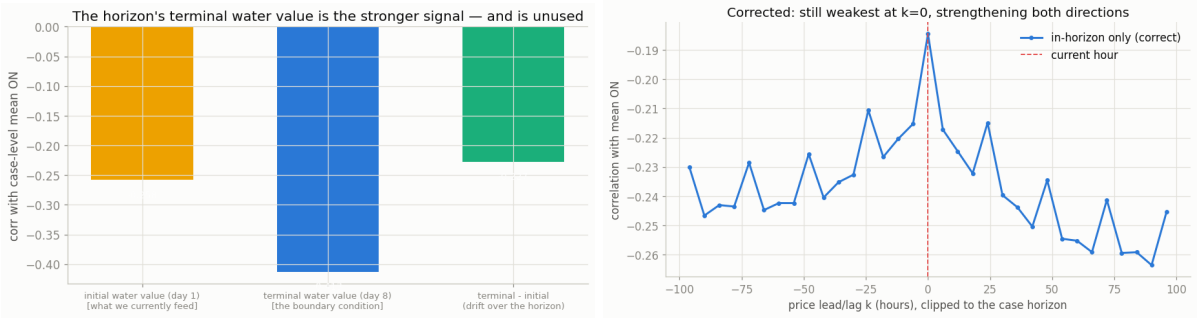


Figure 12: Left: initial and terminal water value against case-level commitment. Right: correlation of commitment with price k hours earlier or later, within the case horizon.

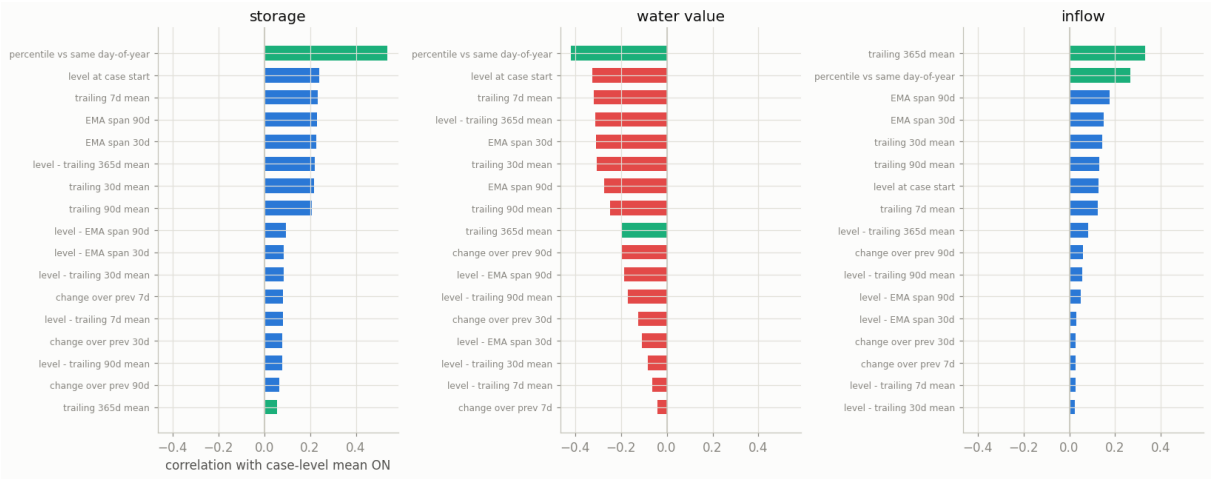


Figure 13: Seventeen backward-looking transformations each of storage, water value and inflow, ranked by correlation with commitment.

2.6 Clean up features

check	action
missing values	none in any core or extended file; reservoir names consistent across files
constant series	storage of Hyljelihyl and Vatjern is fixed at 50% (no records): dropped
leakage	`total_calculation_time` (SHOP's solve time) exists only for labelled cases and describes the solution: excluded from all features
availability	every feature file covers all 725 test cases (price to 2025-12, inflow to 2025-10, storage and water value to 2025-01-01); labels end in 2022, so no feature may use past labels. The terminal water value of the last test case (2024-12-25) is the file's final row
redundant scaling	normalising a single column (e.g. storage by reservoir size) is a no-op for tree models; scaling was only applied where columns are combined
uninformative encodings	a binary "min-flow active" flag is constant for five of six rivers: the level is used instead
post-decision information	end-of-week storage reflects the decisions being predicted. It is legal (the files cover the test period) and was used in Track A; Track B shows it is not needed (Section 4.4)
unit identity	plant names are sanitised for column names (`Vesle Kjela`) and mapped back from the topology for the submission header

Table 3: Data cleaning.

2.7 Training versus test period

We measured which past period resembles the test set instead of assuming it: a classifier separates each half-year's cases from the 2023–24 test cases using case-level inputs (price level, volatility, negative-price share, water value, spread, fill, inflow). 2021-H2 is the hardest to separate (AUC 0.785), then 2022-H2 (0.879); every other half-year is above 0.95 (Figure 14). We report these two *test-like folds* separately in validation.

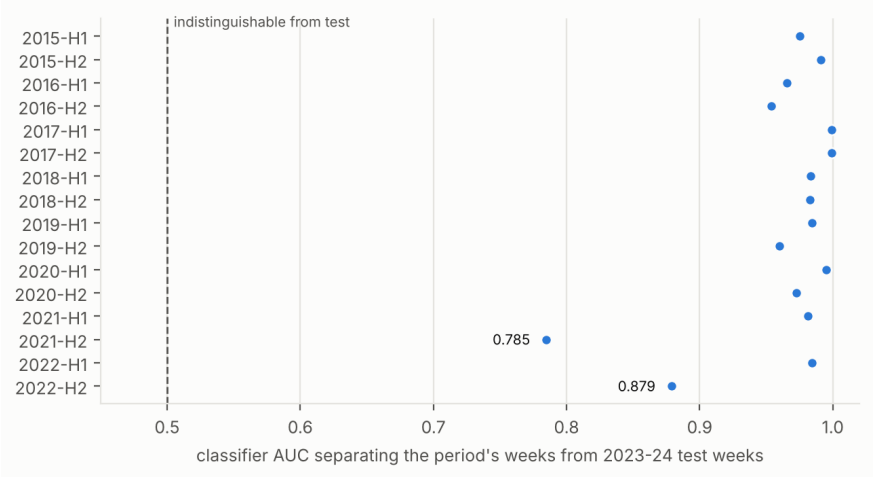


Figure 14: Adversarial validation: how easily each half-year's cases are told apart from the test cases. Lower is more test-like; 0.5 would be indistinguishable.

2.8 What the analysis implied for modelling

finding	consequence
the spread price − water value drives decisions	spread features in Track A; LP margin in Track B
terminal water value is the week's boundary condition	terminal features; the LP's end-of-week value
price matters relative to its week	case-relative price features, never raw levels
slow series confound season	seasonal normalisation (Track A)
units are coupled through the cascade	one joint model for all units; long format
identical units follow a merit order	unit identity and merit rank as features; order constraints in the LP
consecutive cases share 144 hours	temporal validation only (Section 3)
regimes shift; 2021-H2 is most test-like	walk-forward folds with test-like folds reported separately
labels are the output of a known optimiser	Track B: rebuild the optimiser as a feature generator

3 Validation methodology

Random splits leak. A random row split scores 0.957 with the same model that scores 0.835 on a held-out year: rows from one week are near-duplicates, and consecutive weeks share 144 hours at 88% label agreement. Random *case* splits still inflate by +0.105. Every number in this report therefore comes from a temporal split.

Walk-forward. The reference protocol has six half-year folds from 2020-H1 to 2022-H2; each trains on every case starting before the fold and scores the six months after. Earlier stages used a single held-out year (2021, the most test-like year) or four-fold rolling-origin CV; we judged changes on the per-fold sign pattern, since fold-to-fold variation (SD up to 0.06) dwarfs most feature effects. A rotating-block scheme was built and rejected: training on both sides of the held-out block measures interpolation, while the test set requires extrapolation. Experiments train on every third case ("stride 3"); submissions train on all cases.

4 Feature engineering

We used two types of feature engineering: *feature extraction* (new features built from the raw series, Sections 4.1, 4.2 and 4.3) and *feature selection* (Section 4.4).

4.1 Feature extraction I: economic and structural features (Track A)

Twenty feature themes were built and screened, following the EDA's rules: case-relative price, raw spreads, seasonally normalised slow series, a system factor plus reservoir deviations, multi-hop topology for unit-to-reservoir mapping.

family	what it computes	effect
spread	price − water value of the reservoir feeding each plant, positive part, system spread	+0.075
terminal water value	value at the end of the week; price − terminal value; per-plant terminal spreads; slope	+0.034; day fixed: +0.006
merit and capacity	units a spread justifies ($p_{\max}/p_{\min}$), capacity share needed, break-even hours from start costs	in top 15
cascade graph	upstream spreads and water values via multi-hop traversal of the topology	in top 15
count table	per (case, unit): trailing and forward windows of price, inflow, storage; water balance and implied production; unit-relative views	+0.078 (water balance)
hourly structure	the spread summed over windows of several lengths around each hour: a unit on for c hours runs in a high-spread stretch	+0.003
long format	one row per (case, hour, unit): unit-specific features without 14× column cost	+0.006

Table 4: Track A feature families. Effects are the gains measured when each was added (protocols differ between rows).

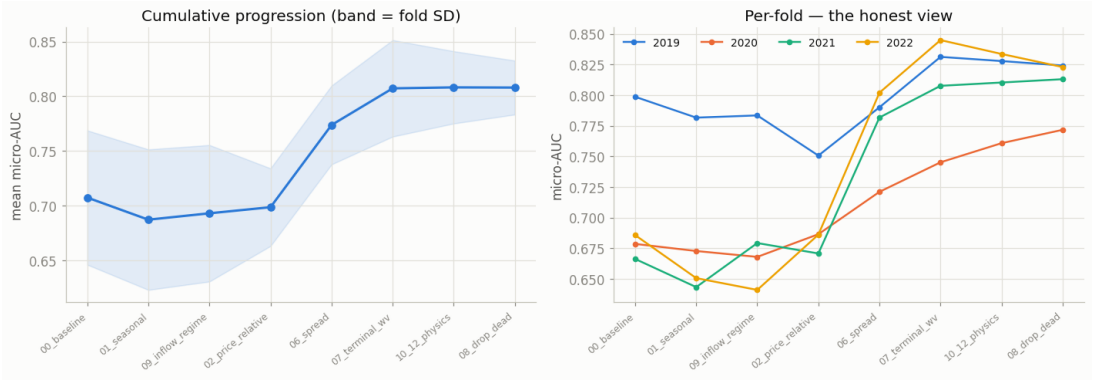


Figure 15: Feature iteration 1: cumulative micro-AUC as stages are added, with the four validation folds (right). The spread and terminal-water-value stages carry the gain; the seasonal stage hurts.

Features that did not help are as informative as the ones that did (Table 5): the strongest univariate correlate in the project, storage's seasonal percentile (+0.54), made the model *worse*.

rejected feature idea	effect
expanding-window seasonal percentile (strongest univariate signal)	−0.020
"physics" stage (heads, MWh per m^3) as features	−0.012
corrected water balance for the four worst units (Songa, Haukeli, Lio, Hogga)	−0.003
each unit's hours allocated as one contiguous block, given the true count	−0.043
day-of-year inflow percentiles from the whole 1958–2025 record	−0.000
min-flow obligation per plant	−0.003
287 slope (rate-of-change) features	−0.002
the count model's full 12-value distribution forwarded; counts of hydraulic neighbours	−0.001

Table 5: Track A feature families that were built, measured and rejected.

4.2 Feature extraction II: a first-principles optimiser as a feature generator (Track B)

Track A approximated SHOP with hand-made features. But the labels are the output of one deterministic optimiser, and every input it used is in our data — the topology file is SHOP-ready. So we rebuilt the optimisation problem itself (`src/shoplite.py`; Short notebook 1) and use its solution as features. The LP is a **feature extractor only**: its output is never submitted; every prediction comes from the learned model.

The linear programme. One per case, the LP relaxation of SHOP's weekly unit-commitment problem, solved with HiGHS (dual simplex; building and solving take 8 s per case on one core on average, 2–25 s for individual cases; SHOP's own recorded solve times are a median 52 s, mean 147 s):

$$\max \sum_t \lambda_t (P_t - L_t) \; - \; \text{start/stop costs} \; - \; \text{river, bypass, spill costs} \; - \; \text{penalties} \; + \sum_r \omega_r V_{r,T}$$

subject to hourly water balance in all 17 reservoirs, the Vest-Vassdraget river junction and the Songa tunnel junction; reservoir bounds; minimum volume and minimum flow (soft, heavily penalised); for every unit an on/off variable $u \in [0, 1]$ carrying the minimum block ($q_{\min}$, $p_{\min}$) plus segments of the concave envelope of its PQ curve at the case's initial head (turbine efficiency interpolated between the reference-head curves, times generator efficiency); start/stop variables with the topology's costs; and main-tunnel and shared-penstock friction L_t as a convex piecewise-linear cubic in plant flow. λ_t is the price and ω_r the terminal value of water. The LP's duals give the hourly shadow price of water in each reservoir, from which each unit's *local water value* (EUR/MWh) and its margin, price − local water value, follow. Simplifications against SHOP: heads frozen at their initial levels, lossless tunnels between reservoirs, and an unknown initial on/off state.

Three conventions measured from the labels. **Terminal water value:** in weeks where a unit switches, its on-hour and off-hour prices bracket the threshold SHOP used; regressing that threshold on terminal water values gives $\approx 1.0\times$ the feeding reservoir's value and $\approx 0\times$ the downstream value (Figure 16; Byrte slope 0.98, R^2 0.99), so one Mm^3 in reservoir r is valued as $\omega_r = \sum_k \text{WV}(\text{reservoir feeding } k) \cdot \text{conv}_k$ over the plants k on its way to the sea. **River flow cost:** `flow_cost_curve` read as total cost per hour (marginal cost = slope) rather than as a marginal cost lifted Kjela's raw AUC from 0.70 to 0.96. **Sibling order:** the merit order from the EDA is imposed as $u_{\text{G1}} \geq u_{\text{G3}} \geq u_{\text{G4}} \geq u_{\text{G2}}$ at Tokke and $u_{\text{G1}} \geq u_{\text{G2}} \geq u_{\text{G3}}$ at Vinje.

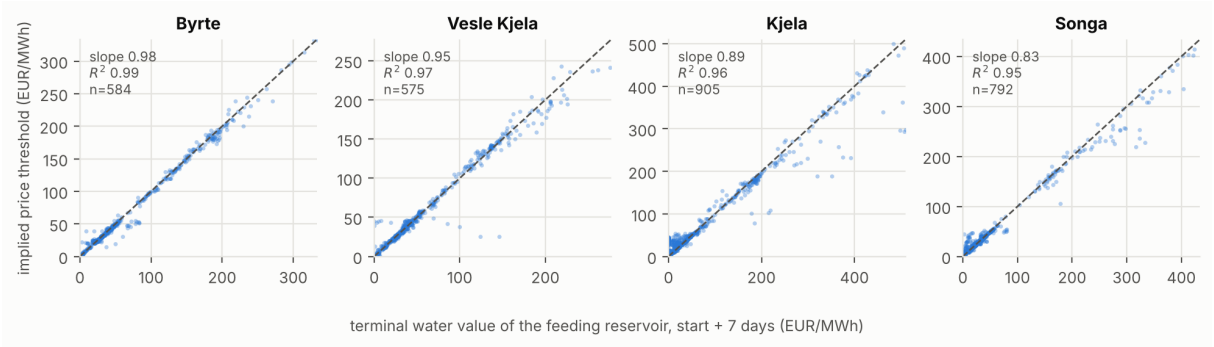


Figure 16: Implied price threshold in every mixed week against the terminal water value of the reservoir feeding the plant. Points lie on $y = x$: each plant breaks even at its own reservoir's water value.

LP versus MILP as a feature. Solving with binary u gave a raw AUC of 0.941 against the relaxation's 0.967 on the same 59-case sample: a 0/1 schedule loses the graded information about how close each hour is to the margin. The relaxation is the better feature.

The 61 features. One row per (case, hour, unit): `lp_` — the LP solution (commitment u, power, flow), its temporal context (rolling windows, run lengths, weekly and daily means, number of units on at the plant), the local water value and margin, and the reservoir trajectories; `ec_` — price level, rank and z-score within the week, spread to the terminal water value, season, inflow; `cn_` — the constraint series; `st_` — the observed storage trajectory; `s_` — unit identity and static attributes.

4.3 Feature extraction III: closed-form physics features for a neural network (Track C)

Track C builds 99 features per (case, unit, hour) for a sequence model (Short notebook 2). The 83 base features follow the same EDA findings as Track A:

group	n	content
economic core	9	margin = price − terminal water value of the intake reservoirs; at the initial value; share of positive hours; daily and rolling (12 h, 24 h min/max) views
price	18	level, within-week z-score and rank, within-day rank and position, daily spread, rolling means, lags and leads of 1/3/6 h, price minus the mean of the remaining hours
water value	9	intake and downstream values, change over the week, daily path and its volatility, a margin against the whole cascade's water value
state	9	fill of the intake, storage and downstream reservoirs, system fill, five large reservoirs
inflow	5	weekly inflow to the plant's storage, relative to its size, system inflow, its anomaly against the 1958–2014 climatology, hourly system inflow
constraints	9	minimum flow on six rivers; distance of three reservoirs to their minimum volume
volume path	8	observed storage changes over the week and implied releases (post-decision information)
unit	8	identity, $p_{\min}$, $p_{\max}$, start cost, position in the plant, number of siblings, head, hour index
calendar	5	local hour and weekday, season, Norwegian holidays
plant context	3	system-mean margin, the unit's margin relative to it, rank across units

Table 6: Track C base features.

Physics features without a solver. Sixteen features are derived in closed form from each generator's efficiency curves at the plant head (minus friction losses, times generator efficiency, reduced to a concave envelope):

- **break-even prices** at minimum, full and marginal load — the intake water value scaled by how much less efficient that operating point is than the plant's best one — and the margins of price over them;
- a **weekly water budget**: hours of full-plant flow the reservoirs feeding the plant can afford (storage above the minimum volume plus inflow), and hours it is *forced* to run because the intake would otherwise overflow;
- a **greedy schedule**: the plant runs in its k best-margin hours, k the number of positive-margin hours clipped between the forced and affordable hours; whether each hour is in it and how far from the cut-off;

- **block economics**: for each run of positive-margin hours, its length and profit at full load against the start cost, and for each negative run, whether idling at $p_{\min}$ is cheaper than stopping and restarting.

These approximate, hour by hour, what Track B's LP computes jointly. They add +0.004 to the yearly cross-validated AUC of the network in its original run (0.9697 → 0.9740; Section 5.4), most of it on 2021 (0.9514 → 0.9658), the most test-like year (Section 2.7).

4.4 Feature selection

Pruning by gain. All 299 Track A features were ranked by gain from one fit and the feature-set size swept: the top 15 beat the full set (0.8157 vs 0.8144 over four folds, with half the fold SD). All 15 survivors are economic quantities; no raw price, storage, inflow or calendar column survived. They are the 15 `PRUNED_FEATURES` of the Track A notebook.

feature set	mean micro-AUC (4 folds)	SD
all 299	0.8144	0.0304
top 15 by gain	**0.8157**	**0.0158**
top 15, 900 trees	0.8097	0.0151

Table 7: Pruning: 15 features beat 299.

Model-based ranking. The count table holds 446 candidate (case, unit) features. A 6-class XGBoost model over hours-on bins is fitted *only to rank them*; the top 120 by importance, plus up to 30 unit-relative features ranked below them, are attached per unit (141 in the submission). Three selection methods were compared; every one that used validation data looked better on the validation year and worse on the folds:

selection method	single year 2021	walk-forward
count-model ranking (chosen without validation data)	0.9651	**0.9638**
permutation importance + unit-relative trimming	**0.9671**	∼0.9609 (3 folds, all worse)
permutation top-60 (circular)	0.9668	not run
held-out inner-split ranking	0.9631	not run

Table 8: Feature-selection methods.

Group selection in Track B. Removing one feature group at a time on the walk-forward folds (Figure 20, Section 7) shows the observed storage trace can be dropped at no cost (0.9910 vs 0.9912): the model needs only information available before the week starts. Without the LP features the same learner falls to 0.9483.

4.5 Inputs used by each final model

input	notebook 1: LP + LightGBM	notebook 2: neural network
price (hourly)	✓	✓
inflow (hourly)	✓	✓ (and its 1958–2014 climatology)
storage (daily)	✓	✓
water value, initial and terminal (daily)	✓	✓ (and the daily path)
minimum volume, minimum flow (extended)	✓	✓
topology and system configuration (extended)	✓ (the full LP)	✓ (efficiency curves, heads, costs)
calendar	season	local hour, weekday, season, holidays

5 Predictors: models and algorithms

5.1 Four predictor types on identical features

To separate the effect of the learner from the effect of the features, four types of predictor were trained on the same 61 features (Section 4.2) and scored on the same six walk-forward folds (stride 3; `src/predictor_comparison.py`):

- **logistic regression** — a linear model: median imputation, standardisation, one-hot unit and plant;
- **random forest** — 200 bagged trees, `min_samples_leaf` 100, $\sqrt{p}$ features per split, a quarter of the rows per tree;
- **XGBoost** — boosted trees with Track A's tuned settings (Section 6);
- **LightGBM** — boosted trees with the submitted model's settings.

fold	logistic regression	random forest	XGBoost	LightGBM
2020-H1	0.9864	0.9915	0.9922	0.9925
2020-H2	0.9776	0.9854	0.9869	0.9866
2021-H1	0.9906	0.9939	0.9947	0.9950
2021-H2 (test-like)	0.9810	0.9893	0.9909	0.9914
2022-H1	0.9754	0.9835	0.9871	0.9862
2022-H2 (test-like)	0.9885	0.9934	0.9952	0.9953
mean	0.9833	0.9895	**0.9912**	**0.9912**
test-like mean	0.9848	0.9913	0.9930	**0.9934**

Table 9: Four predictor types on the same 61 features and the same walk-forward folds (micro ROC-AUC). `report/runs/predictor_comparison_s3.csv`.

The ranking is the same on every fold: the linear model below the bagged forest, the forest below the two boosted ensembles. XGBoost and LightGBM tie (0.9912 each; 0.9930 and 0.9934 on the test-like folds), so the choice of boosting library does not matter; we kept LightGBM, the submitted model. More important, every type scores far above Track A's XGBoost on its own engineered features (0.9619): even logistic regression on the LP features reaches 0.9833. The gain of Track B is in the features the optimiser extracts, not in the learner.

5.2 Every model we trained, helpful or not

Table 10 lists every model we trained, in order. Protocols changed as the project matured (single year 2022 → four-fold rolling origin → single test-like year 2021 → six-fold walk-forward), so scores are comparable only within a protocol.

track	model	protocol	AUC	verdict
A	14 independent XGBoost classifiers, raw features	year 2022	0.647	baseline
A	one multi-output XGBoost tree (shared splits across units)	year 2022	0.687	kept
A	multi-output tree, 300 / 700 trees	year 2022	0.686 / 0.695	300 kept (plateau)
A	multi-output tree + engineered features	4-fold CV	0.808	kept
A	pruned to top 15 features (900 trees: 0.810)	4-fold CV	0.816	kept
A	stacked two-stage model (uniform / gated)	year 2022	0.839 / 0.843	superseded
A	oracle: true hours-on, allocated by spread (upper bound)	year 2022	0.977	diagnostic
A	count model, then allocate hours by spread	year 2022	0.780	rejected
A	three-class count model + allocation	year 2022	0.826	rejected
A	pointwise model + count-budget augmentation	year 2022	0.855	kept
A	+ multi-horizon count features	year 2021	0.856	kept
A	+ water-balance (post-decision) features	year 2021	0.912	kept
A	same without post-decision information	year 2021	0.855	diagnostic
A	+ unit-relative and absolute count features	year 2021	0.917	kept
A	walk-forward estimate of that model	walk-forward	0.902	
A	count features into the pointwise model	year 2021	0.929	kept
A	compact distilled architecture	year 2021	0.940	kept
A	water-balance fixes (v2)	year 2021	0.937	rejected
A	walk-forward estimate, compact model	walk-forward	0.935	
A	binned 6-class count head	year 2021	0.942	kept
A	unit-resolved count model / stationarised variant	year 2021	0.912 / 0.940	rejected
A	+ hourly temporal structure	year 2021	0.944	kept
A	block allocation; per-unit calibration; climatology; min-flow	year 2021	0.906–0.942	rejected
A	long format, one row per (case, hour, unit)	year 2021	0.950	kept
A	long format with Lio up-weighted	year 2021	0.948	rejected
A	terminal water value corrected to start +7 days	year 2021	0.960	kept
A	tuned pointwise model; tuned count model (no change)	year 2021	0.963 / 0.963	kept
A	slope features; richer count outputs; unit masking	year 2021	0.961–0.963	rejected
A	**submission 1**: count model only ranks features	walk-forward	0.9638	submitted
A	**submission 1** on the public leaderboard	test set	**0.98058**	
B	MILP relaxation's schedule as a feature (59 cases)	sample	0.941	rejected
B	LP relaxation's commitment feature alone, no learning	walk-forward	0.9710	diagnostic
B	**LightGBM on LP features (submission 2)**	walk-forward	**0.9912**	selected
B	**submission 2** on the public leaderboard	test set	**0.99522**	
B	LightGBM on LP features, no storage trace / LP only	walk-forward	0.9910 / 0.9897	ablation
B	LightGBM without the LP features	walk-forward	0.9483	ablation
B	LP features: logistic regression / random forest / XGBoost	walk-forward	0.983 / 0.990 / 0.991	comparison
C	UCNet on the 83 base features	yearly CV	0.9697	superseded
C	LightGBM on the 99 base + physics features	yearly CV	0.9722	rejected
C	**UCNet on base + physics features**	yearly CV	**0.9740**	kept
C	**selected run of that model** on the public leaderboard	test set	**0.99024**	selected

Table 10: All models, in the order they were trained. "Year 2022" and "year 2021" hold out one year; 4-fold CV is rolling-origin; walk-forward is six half-year folds; yearly CV is Track C's expanding window (2019–2022).

Terms in the table. A *pointwise* model predicts each unit-hour directly; a *count model* predicts how many hours a unit runs in the week. *Allocation* gives a unit its predicted hours in the hours with the highest spread. A *stacked two-stage* model fits per-unit residual models on top of the joint model; *gated* applies them only to units where they helped on the previous

year. *Count-budget augmentation* adds the difference between the predicted count and each hour's spread rank to the pointwise model's logits. *Multi-horizon* count features are trailing and forward windows. The *compact distilled* architecture passes five count-model outputs per unit to the pointwise model instead of hundreds of raw features, and the *binned head* is a single 6-class model over bins of hours on. The *unit-resolved* variant replaces per-plant columns with the unit's own, upstream and downstream views; the *stationarised* one replaces drifting level features with ratios. *Water-balance fixes* correct the storage accounting of the four worst units (Songa, Haukeli, Lio, Hogga). *Unit masking* blanks a block of features for units outside a chosen set. *Block allocation* forces each unit's hours into one contiguous window.

Three architectural ideas carried Track A. Joint modelling of the correlated units (0.647 → 0.687). A count model over hours-on per unit-week: given the true count, allocating hours by spread reaches 0.977, so the hard part is one number per unit-week; the count model first augmented and later only ranked features, because forwarding its predictions cost 0.008. And the long format, which made unit-specific features affordable.

5.3 Track B: LightGBM on LP features

Both submitted models were run on the same six walk-forward folds on the same machine, each training on every third case; the submitted XGBoost model re-run this way reproduces its recorded 0.9638 to within 0.002.

fold	XGBoost (Track A)	LP feature alone	LP + LightGBM	Δ
2020-H1	0.9535	0.9720	**0.9925**	+0.039
2020-H2	0.9455	0.9671	**0.9866**	+0.041
2021-H1	0.9765	0.9787	**0.9950**	+0.019
2021-H2 (test-like)	0.9504	0.9695	**0.9914**	+0.041
2022-H1	0.9672	0.9586	**0.9862**	+0.019
2022-H2 (test-like)	0.9784	0.9802	**0.9953**	+0.017
mean (SD)	0.9619 (0.014)	0.9710 (0.008)	**0.9912 (0.004)**	+0.029

Table 11: Walk-forward micro ROC-AUC. "LP feature alone" scores the LP's commitment feature u directly, with no learning: it measures the feature, it is not a submitted predictor.

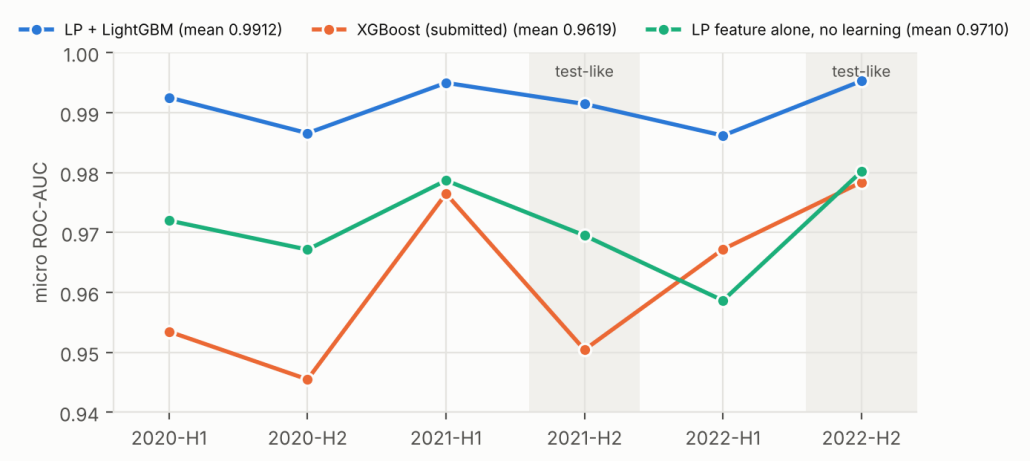


Figure 17: Walk-forward results per fold. The LP commitment feature on its own already ranks hours better than the trained Track A model on five of six folds.

Every fold and every unit improves (Figure 18); the two units Track A never fixed move most

(Lio 0.864 → 0.979, Haukeli 0.920 → 0.973), and the fold SD falls from 0.014 to 0.004.

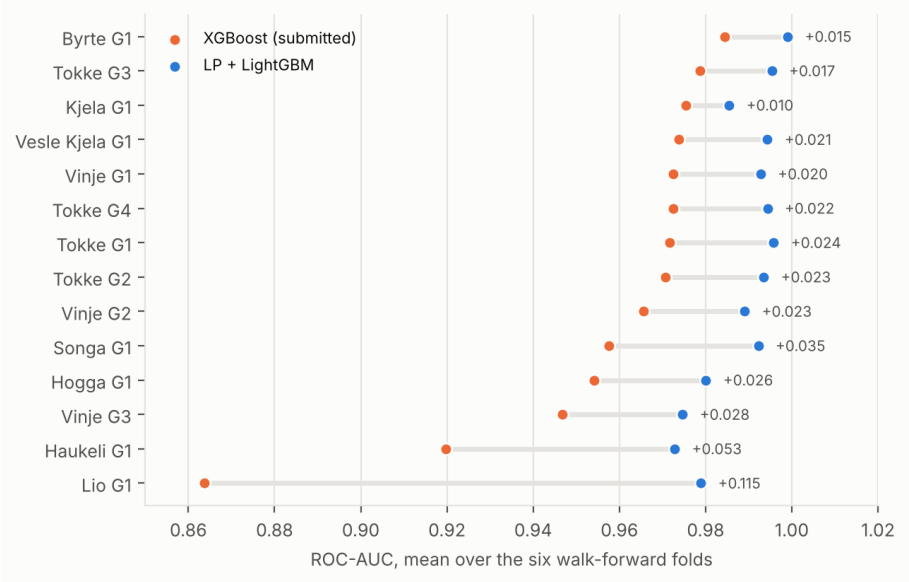


Figure 18: Per-unit ROC-AUC, mean over the six folds, Track A (orange) and Track B (blue).

5.4 Track C: a sequence neural network

Each case is one sample: an input of 14 units × 168 hours × 99 features and a target of 14 × 168 commitments. The network, UCNet (244,161 parameters), mirrors the structure the EDA found:

- a two-layer input projection to width 96, concatenated with a learned 16-dimensional embedding of the unit;
- a residual temporal convolution (two layers, kernel 5, the second dilated) and a residual two-layer bidirectional GRU along the 168 hours of each unit — commitments come in blocks;
- residual multi-head self-attention (4 heads) across the 14 units at each hour — units share water and are coupled through the cascade;
- a two-layer head giving one logit per unit-hour.

Inputs are scaled per feature (median, 10–90% range, clipped at ±8). Training minimises binary cross-entropy over all unit-hours with AdamW and a one-cycle learning-rate schedule, 20 epochs, batches of 32 cases; the submission averages three seeds' probabilities.

Validation. Track C used an expanding window by year: train on cases starting more than 7 days before 1 January of year Y (consecutive cases overlap, so the gap prevents leakage), score year Y. The figures in this subsection are from the network's original run, with one seed; the selected prediction comes from a later run of the same model in Short notebook 2 (Section 8).

validation year	AUC	validation AUC after epoch				
		4	8	12	16	20
2019	0.9837	0.9783	0.9790	0.9817	0.9833	0.9837
2020	0.9685	0.9643	0.9649	0.9641	0.9696	0.9685
2021	0.9658	0.9637	0.9648	0.9626	0.9670	0.9658
2022	0.9779	0.9748	0.9710	0.9772	0.9781	0.9779
mean	**0.9740**					

Table 12: Track C, yearly cross-validation (micro ROC-AUC).

On the same yearly folds, the same network without the 16 physics features scores 0.9697, and LightGBM on the same 99 features 0.9722: the sequence model adds +0.002 over boosted trees on these features, and the physics features +0.004 over the base set. On the public leaderboard the network scores **0.99024**, far above its cross-validation — the same direction as Tracks A and B, whose walk-forward estimates also understated their leaderboard scores (the test period is post-crisis, the folds are not).

The two selected models on the same folds. Track B's walk-forward and Track C's yearly cross-validation use different folds, so their figures are not comparable. We therefore also scored LP + LightGBM, with Short notebook 1's settings, on Track C's protocol: the same yearly folds, the same 7-day gap, and all training cases with no stride (`src/lp_yearly_cv.py`).

validation year	Track C (network)	Track B (LP + LightGBM)	difference
2019	0.9837	**0.9927**	+0.009
2020	0.9685	**0.9905**	+0.022
2021	0.9658	**0.9939**	+0.028
2022	0.9779	**0.9914**	+0.013
mean	0.9740	**0.9921**	+0.018

Table 13: The two selected models on Track C's yearly folds (micro ROC-AUC). Track C's figures are from a single seed of its original run.

Track B is ahead in every year. Its lead is largest in 2021 (+0.028), the year that most resembles the test period (Section 2.7), and its scores vary less across years (0.9905–0.9939, against 0.9658–0.9837). The public leaderboard points the same way, by a smaller margin (0.99522 against 0.99024), which is why Short notebook 1 holds our best score.

6 Hyperparameters: how the values in the notebooks were obtained

The course allows a few hyperparameter values to be set directly in the short notebooks, provided the report says how they were obtained. Every value set by hand in either notebook is listed here.

6.1 Track A: XGBoost (first submission, not selected)

Track A's notebook is no longer one of the two selected predictions, but its values are documented for completeness. The point model's and count model's settings were checked by coordinate descent, one knob at a time, on the most test-like validation year (2021) at stride 3.

parameter	value	how it was obtained
`min_child_weight`	500	raised from 1: +0.0018 (0.9604 → 0.9622). Predicted from the data: hourly rows are autocorrelated at 0.995, so the effective sample size is ∼50× below the row count and small leaves fit single blocks
learning rate × rounds	0.05 × 600	from 0.10 × 300: +0.0010 (0.9622 → 0.9632)
`max_depth`	8	inherited; 10 and 12 gave 0.9621 and 0.9620: no gain
`colsample_bytree`, `subsample`	0.8, 0.8	inherited; column sampling 0.5 and 0.3 gave 0.9625 and 0.9626: no gain
`max_bin`	256	library default
count model	400 trees, rate 0.05, depth 5, `colsample` 0.6	set when the count model was built; screened on log-loss (`min_child_weight` 10/50/200, depth 4/6/8, 800 and 1,200 trees): nothing beat it
forwarded count features	top 120	fixed when the long format was built; 180 and 200 gave 0.9609 and 0.9618 against 0.9632
unit-relative extras	up to 30	set by hand, not tuned: unit-relative and absolute features together beat either alone
`PRUNED_FEATURES`	15 names	gain ranking and size sweep (Section 4.4)
seed	0	fixed for reproducibility

Table 14: Hyperparameters of the Track A notebook.

6.2 Short notebook 1: LightGBM on LP features (Track B)

These settings were *not* tuned for this model before it was submitted: they were carried over from Track A's tuned XGBoost, translated to LightGBM's parameter names.

parameter	value	how it was obtained
learning rate, rounds	0.05, 600	as Track A
`min_data_in_leaf`	500	as Track A's `min_child_weight`, for the same autocorrelation reason
`num_leaves`	127	LightGBM grows trees leaf-wise; 127 leaves is one fewer than a full depth-7 tree, the capacity range Track A found sufficient (depth 8; deeper gave nothing)
`feature_fraction`, `bagging_fraction`	0.8, 0.8	as Track A's column and row sampling; bagging every iteration
`lambda_l2`	1	XGBoost's default L2 penalty, kept for parity
`max_bin`	255	library default
categorical features	unit, plant	native categorical splits instead of one-hot
seed, threads	0, 4	reproducibility; the course's 4-core limit

Table 15: Hyperparameters of Short notebook 1.

Sensitivity, checked after submission. Because these values were inherited rather than tuned, we varied each one around its submitted value, one at a time, on the two test-like walk-forward folds (stride 3, as in Table 11; `src/lgbm_sensitivity.py`):

variant	2021-H2	2022-H2	mean	vs. submitted
submitted	0.9914	0.9955	**0.9935**	—
`num_leaves` 31	0.9912	0.9951	0.9932	−0.00029
`num_leaves` 255	0.9915	0.9954	0.9934	−0.00001
`min_data_in_leaf` 100	0.9910	0.9956	0.9933	−0.00017
`min_data_in_leaf` 2000	0.9914	0.9953	0.9933	−0.00012
learning rate 0.1, 300 rounds	0.9909	0.9953	0.9931	−0.00035
learning rate 0.025, 1,200 rounds	0.9916	0.9954	0.9935	+0.00004
`feature_fraction` 0.5	0.9917	0.9953	0.9935	+0.00005
`feature_fraction` 1.0	0.9910	0.9953	0.9931	−0.00032

Table 16: One-at-a-time sensitivity of Short notebook 1's LightGBM settings (micro ROC-AUC). This run was on a different machine from Table 11, which is why the submitted settings score 0.9955 rather than 0.9953 on 2022-H2.

No change moves the mean by more than 0.0004. Two variants score above the submitted settings, `feature_fraction` 0.5 and the slower learning rate, both by less than 0.0001 and on only one of the two folds (2021-H2). The submitted model is therefore not sitting on a sharp optimum that it happened to inherit from Track A: within this range, LightGBM's settings hardly matter, and the score comes from the LP features (Section 4.4). We did not change the submitted model after seeing these results.

6.3 Short notebook 2: UCNet (Track C)

The network's architecture and optimiser settings were set by hand during development; no record of a systematic search over them was kept. The table gives each value and the reasoning it follows; the number of epochs is the one value checked against validation curves.

parameter	value	how it was obtained
epochs	20, last-epoch weights	the per-epoch validation curves (Table 12) still rise or flatten at epochs 16–20 in every year, so training runs a fixed 20 epochs with no early stopping
hidden width, unit embedding, attention heads	96, 16, 4	set by hand during development; keeps the network small (244,161 parameters) next to ∼41,000 labelled unit-weeks
learning rate and schedule	$2 \cdot 10^{-3}$, one-cycle, 15% warm-up	set by hand; standard settings for AdamW on a model of this size
weight decay, dropout, gradient clipping	10^{-2}, 0.1, norm 1.0	set by hand; conventional regularisation for a recurrent network
batch size	32 cases	set by hand; one case is already 14×168 outputs
input scaling	median, 10–90% range, clip ±8	robust to heavy tails: prices reach 898 EUR/MWh and margins swing by hundreds within a week
seeds	0, 1, 2; mean of probabilities	averaging seeds reduces the variance of a single network

Table 17: Hyperparameters of Short notebook 2.

Constants of the LP. The optimiser in Short notebook 1 also contains fixed numbers that are model constants rather than learning hyperparameters: unit conversions ($K = 0.0036\,\mathrm{Mm^3}$ per $\mathrm{m^3/s}$ for one hour; $P = 9.81 \cdot 10^{-3}\,HQ\eta\,\mathrm{MW}$); deficit penalties of 5,000 EUR per $\mathrm{m^3/s{\cdot}h}$ of

missing minimum flow and 10^5 EUR per Mm^3 of missing minimum volume, set by hand so that the constraints are expensive to break but every case stays feasible (not tuned); and the three conventions measured from the labels (Section 4.2). All physical data (curves, costs, limits) is read from the topology file.

7 Model interpretation

7.1 Track A: what the trees learned

The pruned 15 features are all economic quantities — spreads, margins, break-even hours, the share of capacity a spread justifies — with topology used only to pick *which* water value and *which* capacity; no raw price, storage, inflow or calendar column survived. The single most important feature of the early model, by a factor of 14, was the binary "price > system water value": SHOP's dispatch condition, rediscovered (Figure 19). In the count model the top feature was the system storage change over the week — released water, i.e. production — and the binding-constraint term min(hours of water, hours profitable). Hourly-varying features held 96% of the gain: micro-AUC rewards ranking hours *within* a week, which a case-level constant cannot do.

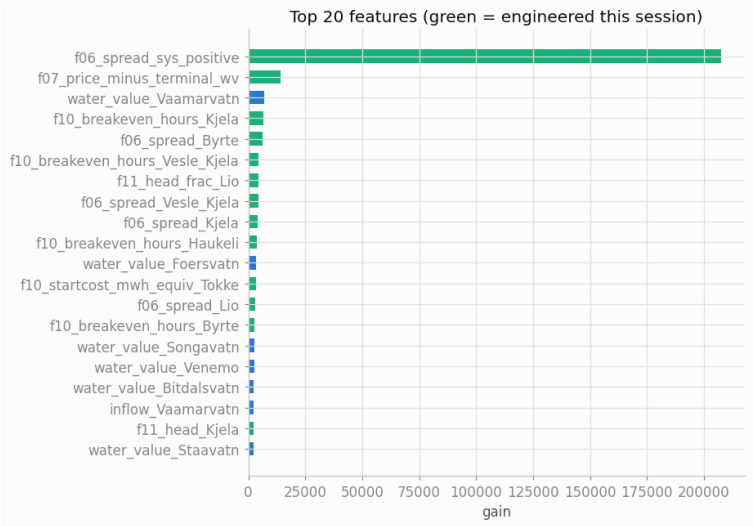


Figure 19: Track A, first feature iteration: gain importance. The binary spread sign — SHOP's own dispatch condition — dominates.

7.2 Track B: the model leans on the optimiser

Ablation. Removing one feature group at a time on the same folds (Figure 20): without the LP features the same learner scores 0.9483 — *below* Track A — so the gain is neither LightGBM nor the feature engineering. The LP features plus unit identity alone reach 0.9897. Removing the observed storage trajectory changes nothing (0.9910 vs 0.9912): the model uses only information available before the week starts, unlike Track A's best features.

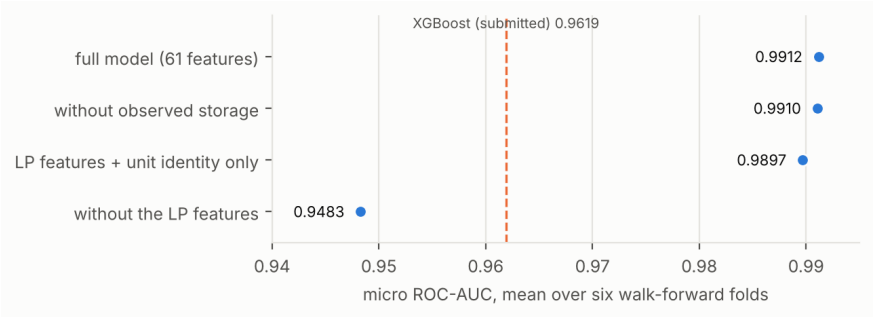


Figure 20: Ablation: mean walk-forward micro-AUC with one feature group removed.

Importance. The LP commitment and flow alone carry 71% of the split gain, LP features 86% in total (Figure 21). Unit identity (5%) is how the model learns per-unit corrections — e.g. that SHOP runs Lio harder than our LP does.

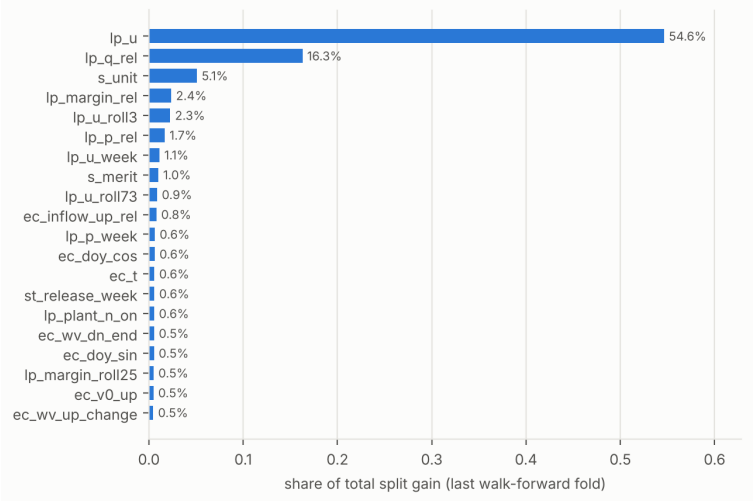


Figure 21: LP + LightGBM: share of split gain, top 20 features (last walk-forward fold).

Contributions. Split gain counts how much a feature is used in training, not how much it moves a prediction. TreeSHAP contributions (LightGBM's `pred_contrib`) on 20,000 validation rows of the test-like 2021-H2 fold (fold AUC 0.9914) give the per-prediction view (Figure 22). The LP commitment u moves the log-odds by 2.05 on average, nearly twice any other feature. Unit identity is second (1.14): it is the channel for per-unit corrections. Third is the margin, price minus the local water value from the LP duals (0.93), with its 7-hour rolling mean close behind: the learner reads how far each hour sits from the dispatch threshold, exactly the graded information a binary schedule discards (Section 4.2). By group, LP features carry 70% of the total absolute contribution, economics 15%, unit identity and static attributes 13%, the observed storage trace 2% and the constraint series 1%. Economics weigh more here than in split gain (15% against 6% of the gain in the last fold): many small adjustments rather than a few decisive splits.

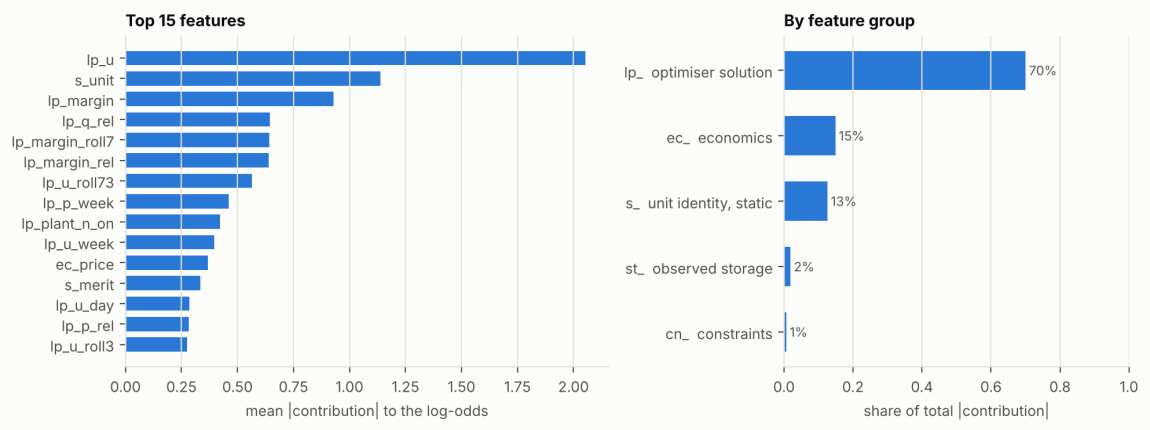


Figure 22: LP + LightGBM, 2021-H2 fold: mean absolute TreeSHAP contribution to the log-odds, top 15 features (left) and share by feature group (right).

How the learned layer reads the LP. The relaxation's u is already roughly a probability (Figure 23): SHOP commits in about 1% of hours where $u = 0$ and 94% where $u = 1$, and the on-rate rises close to the diagonal in between. The departures are unit-specific, which is why unit identity ranks second. Lio, which SHOP runs to meet the Vest-Vassdraget minimum flow, is committed in every hour where the LP gives it $u \geq 0.6$ and in three quarters of hours at 0.4–0.6. Hogga, whose timing is degenerate in our LP when Bandak must fill, is committed in only 45% of hours at u between 0.6 and 0.8.

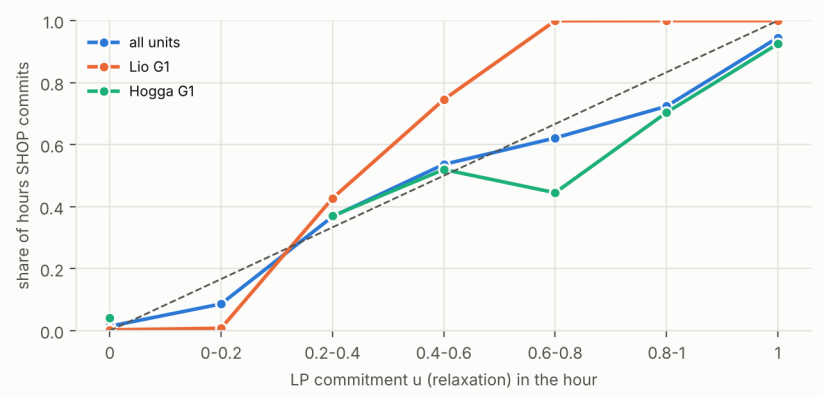


Figure 23: Share of hours SHOP commits, by the LP relaxation's u in that hour (2021-H2 fold). Dashed: perfect calibration.

One week. Figure 24 shows a test-like case the LP finds harder than three quarters of its fold. The relaxation spreads Tokke's commitment fractionally over four identical units and holds Lio at $u \approx 0.45$ until day 3.5. The learned layer concentrates Tokke on G1, the unit SHOP picks first, and sharpens Lio into the on-block SHOP chose; its within-case AUC is 0.999 against the LP's 0.974. Its remaining error is inherited from the LP: Kjela, on all week in SHOP's schedule, dips where the relaxation dips (days 4.4–5.6).

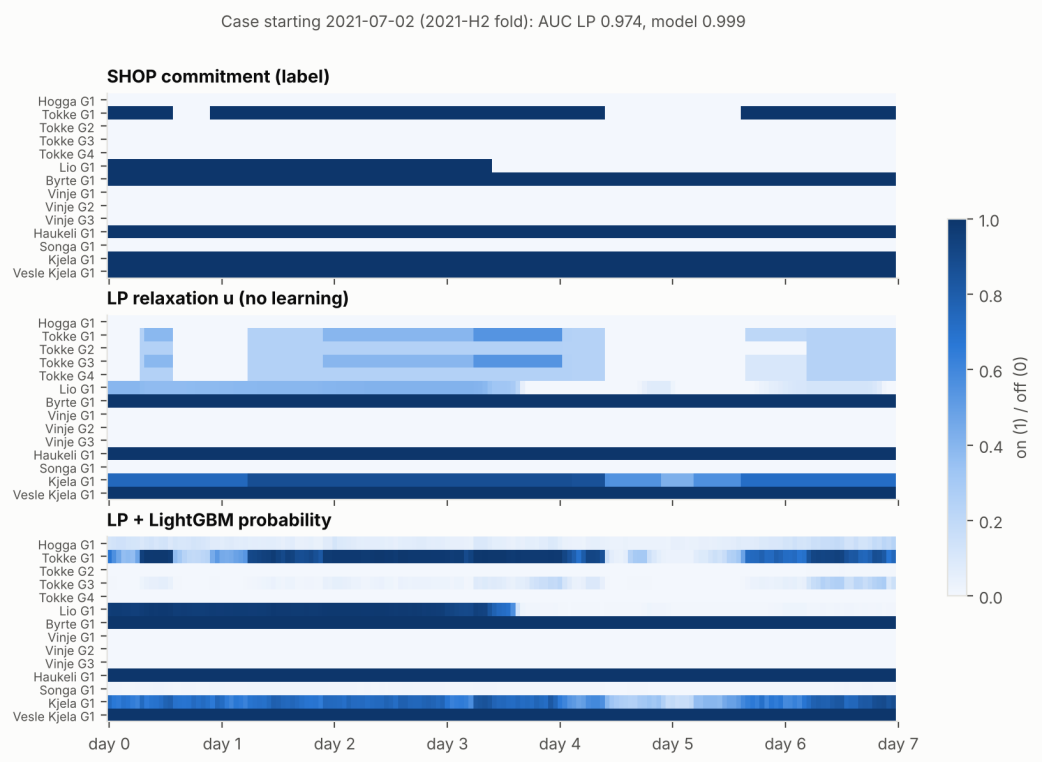


Figure 24: Case starting 2021-07-02: SHOP's commitment (top), the LP relaxation with no learning (middle) and the LP + LightGBM probability (bottom), one row per unit, one column per hour.

The optimiser itself is interpretable. Its decisions are explained by quantities with physical meaning: each unit runs when the price exceeds its local water value, which equals its reservoir's terminal water value unless a constraint binds — a full reservoir lowers it (water would otherwise spill), a minimum-flow obligation can force a unit on below it, an empty reservoir raises it. The three conventions of Section 4.2 are themselves findings about how the labels were produced.

7.3 Track C: what the physics features contribute

The network's own ablation (Section 4.3) is its interpretation result: removing the 16 physics features costs 0.004 AUC overall and 0.014 on 2021, the year that most resembles the test period. The features that carry the gain are the ones that encode the same economics as Track B's LP — break-even prices from the efficiency curves, the weekly water budget and block profit against start costs — computed per plant in closed form instead of jointly by a solver.

8 The two selected predictions

	Short notebook 1	Short notebook 2
model	LightGBM on LP features (Track B)	sequence neural network (Track C)
Kaggle public score	0.99522	0.99024
validation	walk-forward 0.9912 (6 half-year folds)	yearly CV 0.9740 (2019–2022, original run)
trains on	all 2,922 labelled cases	all 2,922 labelled cases, 3 seeds
runtime, 4 CPU cores	∼2 h (3,647 LPs) + 6 min	∼2 h (1.9 min per epoch); 32 min measured on an Apple M4 GPU
peak memory	∼5 GB	∼8 GB
reproduces the uploaded file	to $2 \cdot 10^{-13}$	the selected file is this notebook's saved run; features are exact, training is not bit-reproducible (two runs correlate 0.9993)

Table 18: The two predictions selected on Kaggle. Both notebooks start from the raw data, run offline and stay within the course limits (12 hours, 4 cores, 32 GB).

9 Discussion

Why Track B works. Track A spent most of its effort recovering, through features, a decision process that is written down: the spread is the dispatch condition, the count is the weekly water budget, the block structure comes from start costs, the merit order from penstocks. Solving the actual problem gets all of these at once, consistently across units and reservoirs, and extrapolates to unseen price regimes (negative prices, the post-crisis level) from physics rather than from training data. The learner is left with a smaller, more stationary job: the residual between our simplified replica and SHOP.

Relation to the task's purpose. InterOpt wants ML to speed up the optimiser. Our best model still solves an LP, but one roughly 6–18× cheaper than SHOP's recorded solve times, and its output is a calibrated commitment probability that could warm-start SHOP's MIP. Unlike Track A's best features, it needs no post-decision information.

Limitations and next steps. The remaining errors are fidelity gaps of the replica: heads frozen at their initial level, Lio's min-flow behaviour, weeks where SHOP runs units below their water value (e.g. 2019-07-09), and degenerate weeks where a reservoir must fill and the timing of release is arbitrary. Each fix to the replica feeds the learned layer directly. Three conventions were read off label statistics that include the validation years; they are structural and stable across years, but the protocol is not blind to them.

What we would do differently. Read the topology file on day one. Track A's three largest late gains were corrections found by reading the documentation; Track B is the same lesson taken to its conclusion.

A Reproducibility

- **Configuration.** `requirements.txt` next to the notebooks pins the package versions both notebooks were run with (Short notebook 2 adds PyTorch). The notebooks contain all code, start from the raw competition files (`data/kernel`, `data/extended`,

`sample_submission.csv`, `prediction_mapping.csv`), need no network and write only intermediate results (the LP cache) and the submission file.

- **Short notebook 1** (LP + LightGBM): solves the 3,647 LPs in parallel (∼2 h on 4 cores, cached after the first run), builds features, trains on all labelled cases and writes the submission (∼6 min, peak ∼5 GB RAM). Executed end to end from the raw data, it reproduces the uploaded file to $2 \cdot 10^{-13}$, and its LP solutions are byte-identical to the stored ones.
- **Short notebook 2** (neural network): builds the 99 features from the raw files (about 1–2 minutes), checks them against values recorded from the original run, and trains three seeds on all labelled cases. Its 99 features match the original run's checksums (mean and SD of every feature on train and test) to the printed precision. The selected file is the output of the executed notebook, run end to end on an Apple M4 GPU (PyTorch 2.15, MPS backend): features in 16 s, the three seeds trained in 32 min (27–54 s per epoch). Training is not bit-reproducible, so a re-run gives a prediction close to, not identical with, that file. The notebook's last cell shows the size of this difference against an earlier upload of the same model: correlation 0.9993, 99.4% of cells on the same side of 0.5, largest single difference 0.33. The total of 84 min printed in the executed notebook includes about 52 min during which the machine was asleep, visible as three slow epochs of seed 1. The notebook uses CUDA, then Apple MPS, then the CPU; no GPU is needed: on 4 CPU cores training takes about 2 h (1.9 min per epoch for 3 seeds × 20 epochs).
- **Track A notebook** (`notebooks/Track A XGBoost (0.98058).ipynb`, not selected): rebuilds our first submission from the raw files (about 40 min on 4 cores; peak ∼24 GB of memory including swap). The uploaded file was produced on an Apple Silicon Mac; on x86 Linux with the same package versions the count model selects 143 count features instead of 141, and the output correlates 0.995 with the uploaded file.
- Source modules, every experiment and its result: `src/`, `report/leaderboard.md`, `report/runs/`.